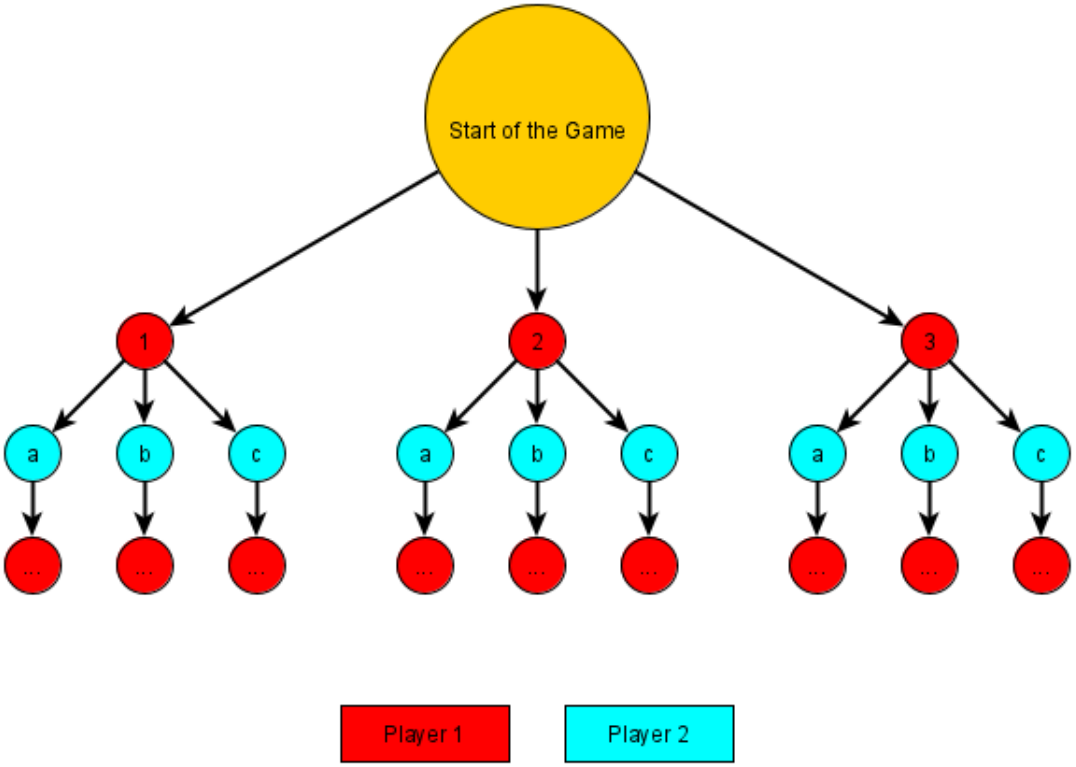
Lecture 28 (Sequential Games)

May 3, 2026

Game Trees

In a sequential game, enumeration of the strategies each player can take is complicated. As such, we often use **game trees** to represent the possible moves a player can take. In a game tree, each path down the tree represents a possible game. Each node represents an action that a player can make in response to the previous player's action. In order to keep the turn orders in place, we view this as a rooted tree, where any leaf node represents the last move of the game.



Each strategy should detail what actions a given player will make in response to the opponent's previous move. As there may be a large number of possible moves, the number of strategies can get quite large with just a few turns. In practice, we often use heuristics and other tricks to find near optimal strategies and to describe strategies.

Poker

One famous application of game theory is to the game of poker. Game theory was initially developed so von Neumann could improve his poker game. Many advances in the theory have stemmed from the study of poker. There are many common variations of poker, and most of them are no where near solved. Some of the most common variants are listed below.

1. Texas Hold 'Em (Both limit and no limit)
2. 5 Card Draw
3. Pot Limit Omaha

4. 7 Card Stud

When poker is played between two players, it is called "Head's Up". Of the variants listed above, only Head's Up Limit Hold 'Em is "solved". In every other variant, the top players can still hold their own against the computer Head's Up. For games involving more than 2 players, the game is no where near solved. Part of what makes poker complicated is the sheer number of card combinations. There are 52 cards in a deck of cards, so there are 52! ways to shuffle the deck. Even if an entire deck isn't dealt throughout a game, the sheer number of strategies (due to bet sizing and hand types) blows up. In fact, the number of strategies (even for simple variations) is far larger than the number of atoms in the observable universe!

Poker Strategy

Since poker is a game of incomplete information, the game becomes quite interesting. In particular, some strategies encourage players to "bluff". This is when a player raises to represent a stronger hand than they actually have. As a game of incomplete information, players can do well by using a combination of two types of strategies, **Exploitative** and **Game Theory Optimal (GTO)** strategies.

Exploitative Strategies

These strategies focus on finding an opponent's mistakes and exploiting them. As multiple hands are played, a good poker player will notice their opponent's tendencies and tells. These will give a good player more information and can lead to an edge. Some of these strategies involve human psychology. Some of them may notice habits from previous moves. In practice these strategies work best against weak players (in fact, if you are not good at poker, this is how a good player will take all your money). We will not focus on these strategies.

GTO Strategies

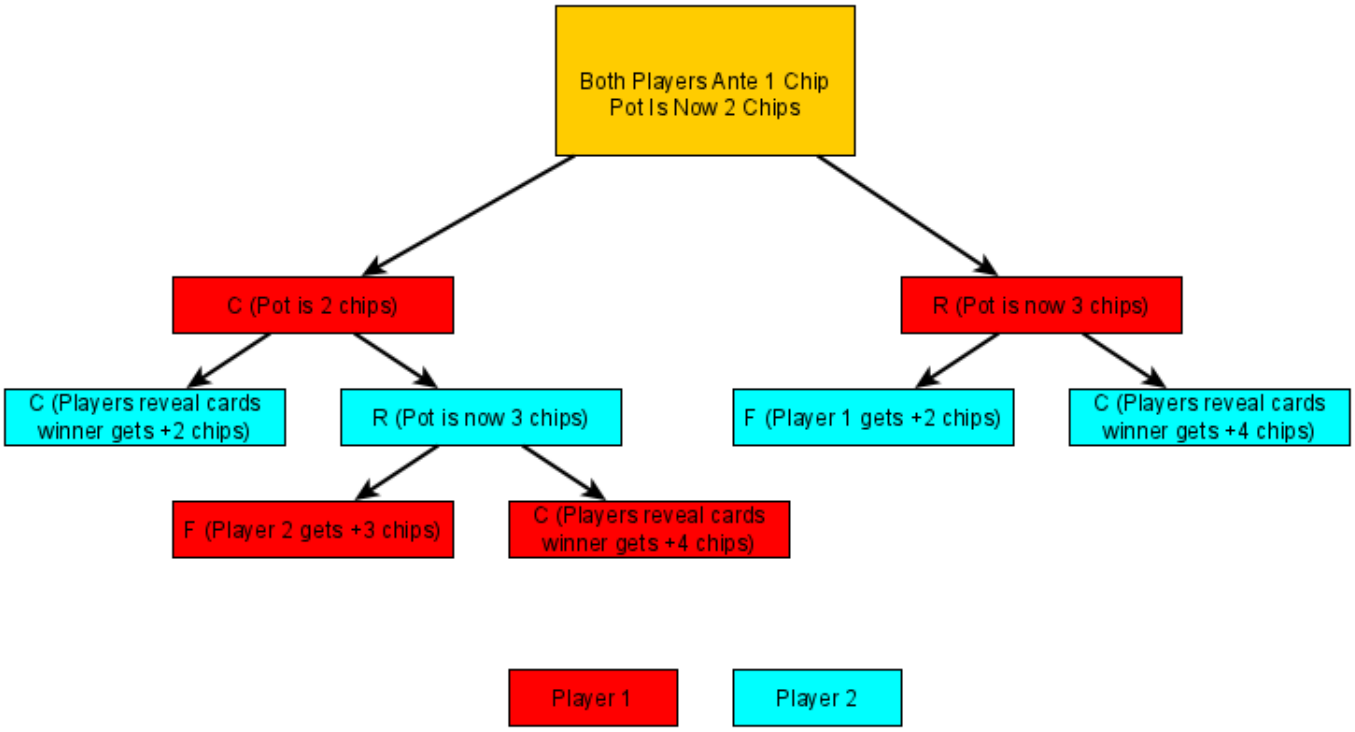
These strategies have become possible in recent years due to increases in computing power and better algorithms. These strategies focus on playing a strategy that isn't exploitable. In practice, these strategies are great against good players, but against weak players, they will not be the best strategy. Professional poker players (especially tournament players) use these types of strategies at the highest levels. These strategies are very mathematical and are mixed strategies.

Kuhn Poker

We will solve the simplest version of poker, Kuhn poker. Kuhn poker was developed by the mathematician Harold W. Kuhn as a simplification of standard poker. This game was solved by Kuhn in the 50s. In this version of poker, the deck consists of 3 cards, a Jack (J), a Queen (Q), and a King (K). These cards are ordered from low to high as

$$\text{J} < \text{Q} < \text{K}$$

To start, each player puts one chip in the pot. After this each player is dealt one card at random from the three cards. A player may take one of four options (with restrictions). A player may Raise (R). This means the player will put one additional chip in the pot. A player may Check (C) or Call (C). If no player has raised yet, a player can check. This means the given player will pass and wait. If another player has raised, then the second player can call the raise. The calling player will also put one additional chip into the pot. If a player raises, the next player can also Fold (F). The folding player will forfeit the hand but will not need to put any additional chips into the pot. After all the actions are completed, if no player has folded, then both cards are revealed. The player with the higher hand win all the chips in the pot. If one player folds, the other player takes all the chips in the pot without having to reveal cards. In Kuhn poker, there can be at most one raise per hand. The possible scenarios are detailed in the game tree below:



If we want to describe each player's strategy (for a given dealt card), we can use two characters. For player 1, we can have the first character represent the first action C or R. The second character will be an X if player one has no remaining action to take. The second character represents player 1's second action in the event player 2 raises. This might be F or C. This yields three possible strategies for player 1 for a given hand. These are CC, CF, and RX. Similarly, we can enumerate player 2's strategies. The first character will represent player 2's response to a C from player 1. The seconds character will represent player 2's response to a R from player 1. This yields four possible strategies for each hand. These are CF, CC, RF, RC. Since each dealt card card may have a different strategy, this yields a total of

$$(3)^3 = 27$$

strategies for player one (as player 1 has to pick one strategy for each possible dealt card). Similarly, player 2 has

$$(4)^3 = 64$$

possible strategies. Using the LP we described in the previous lecture, we can compute the optimal strategy for each player. I claim the expected payoff is $-\frac{1}{18}$. This gives a slight edge to player two. One possible optimal mixed strategy for each player is given below:

	Player 1	
Strategy	CFCCCC	CFCFCC
Percentage	$\frac{1}{3}$	$\frac{2}{3}$

	Player 2	
Strategy	CFCFRC	RFCCRC
Percentage	$\frac{2}{3}$	$\frac{1}{3}$

These were computed in the Python notebook (on Canvas).